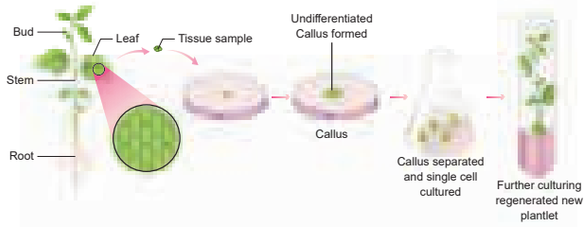
5. Which of the following statement is correct
   a) Agar is not extracted from marine algae such as seaweeds.
   b) Callus undergoes differentiation and produces somatic embryoids.
   c) Surface sterilization of explants is done by using mercuric bromide
   d) P^H of the culture medium is 5.0 to 6.0
6. Select the incorrect statement from given statement
   a) A tonic used for cardiac arrest is obtained from Digitalis purpuria
   b) Medicine used to treat Rheumatic pain is extracted from Capsicum annum
   c) An anti malarial drug is isolated from Cinchona officinalis.
   d) Anti-cancinogenic property is not seen in Catharanthus roseus.
7. Virus free plants are developed from
   a) Organ culture
   b) Meristem culture
   c) Protoplast culture
   d) Cell suspension culture
8. The prevention of large scale loss of biological interity
   a) Biopatent b) Bioethics
   c) Biosafety d) Biofuel
9. Cryopreservation means it is a process to preserve plant cells, tissues or organs
   a) at very low temperature by using ether.
   b) at very high temperature by using liquid nitrogen
   c) at very low temperature of -196 by using liquid nitrogen
   d) at very low temperature by using liquid nitrogen
10. Solidifying agent used in plant tissue culture is
    a) Nicotinic acid b) Cobaltous chloride
    c) EDTA d) Agar
11. What is the name of the process given below? Write its 4 types.



12. How will you avoid the growing of microbes in nutrient medium during culture process? What are the techniques used to remove the microbes?
13. Write the various steps involved in cell suspension culture.
14. What do you mean Embryoids? Write its application.
15. Give the examples for micro propagation performed plants .
16. Explain the basic concepts involved in plant tissue culture.
17. Based on the material used, how will you classify the culture technology? Explain it.
18. Give an account on Cryopreservation.
19. What do you know about Germplasm conservation. Describe it.
20. Write the protocol for artificial seed preparation.

Glossary

Aseptic condition: Preparation of materials free from microbes in *in vitro* cultures.

Cell Culture: Growing of cells *in vitro*, including the culture of single cells or small aggregates of cells in a liquid medium.

Chemically defined medium: A nutritive medium used for culturing cells or tissue; each chemical of this medium is known and defined;

Cybrid: Cytoplasmic hybrid obtained by the fusion of cytoplasm of cells of different parental sources; a term applied to the fusion of cytoplasms of two different protoplasts;

Organogenesis: The process of initiation and development of shoot or root though *in vitro* culture particularly from callus

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